Logistic Regression

Classification

Classification

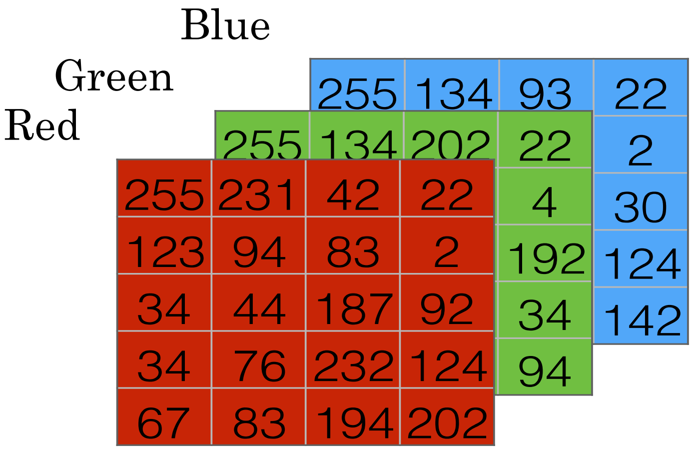
Email: Spam / Not Spam?
Online Transactions: Fraudulent (Yes / No)?
Tumor: Malignant / Benign ?

$y \in \{0, 1\}$

0: “Negative Class” (e.g., benign tumor)
1: “Positive Class” (e.g., malignant tumor)

Binary Classification

$\longrightarrow$ 1 (cat) vs 0 (non cat)



Classification: y = 0 or 1

$h_\theta(x)$ can be > 1 or < 0

Logistic Regression: $0 \leq h_\theta(x) \leq 1$

Logistic Regression

Hypothesis Representation

(How the function will look like?)

Logistic Regression Model

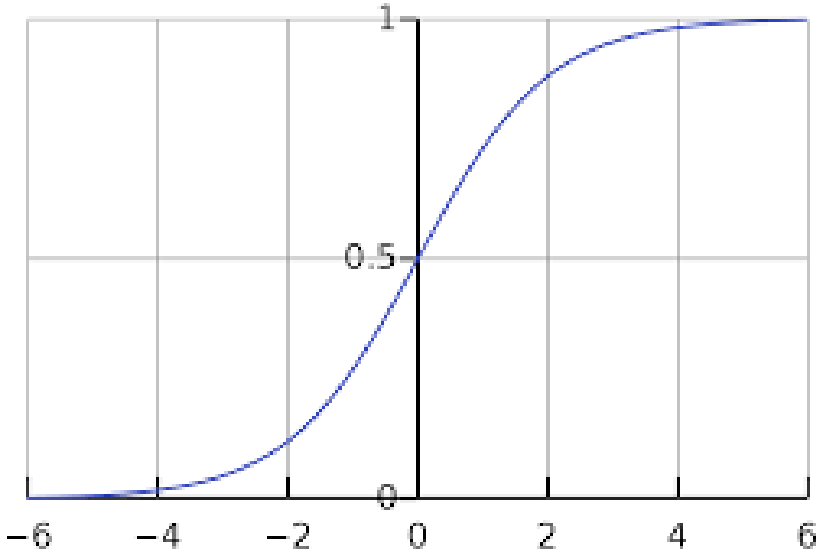
Want $0 \leq h_\theta(x) \leq 1$

$$h_\theta(x) = \theta^T x$$

$$h_\theta(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1+e^{-z}}$$

Sigmoid function
Logistic function



Interpretation of Hypothesis Output

$h_\theta(x)$ = estimated probability that y = 1 on input x

Example: If $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$

$$h_\theta(x) = 0.7$$

Tell patient that 70% chance of tumor being malignant

"probability that y = 1, given x, parameterized by θ"

$$P(y = 0|x;\theta) + P(y = 1|x;\theta) = 1$$
$$P(y = 0|x;\theta) = 1 - P(y = 1|x;\theta)$$

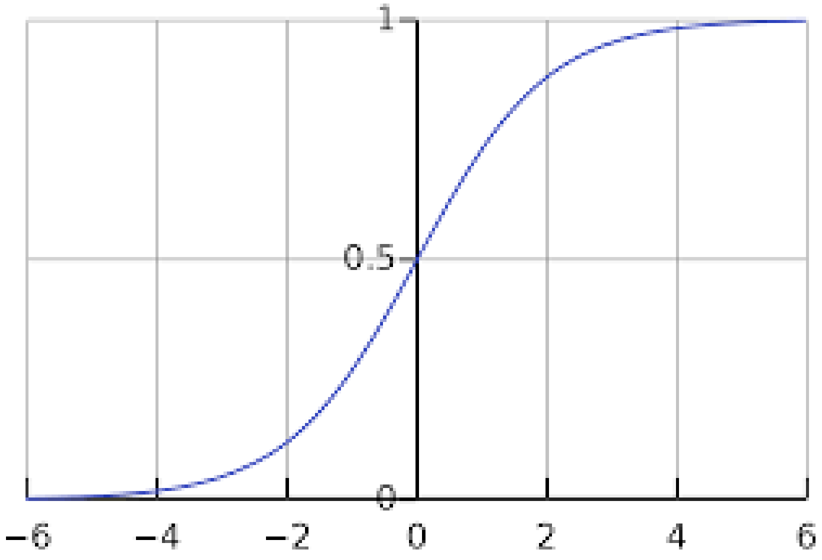
Logistic Regression

Decision Boundary

(Talking about the decision boundary will give us a better sense of what the logistic regression hypothesis function is computing.)

Logistic regression

$$h_\theta(x) = g(\theta^T x)$$
$$g(z) = \frac{1}{1+e^{-z}}$$

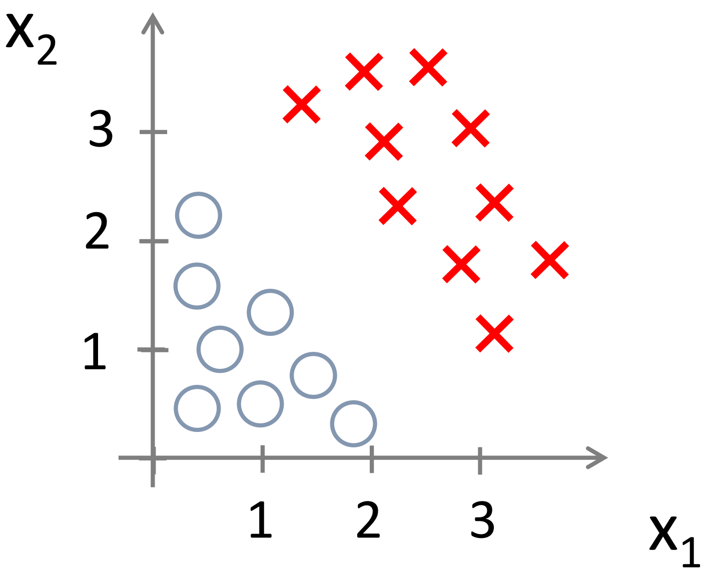


Suppose predict "$y = 1$" if $h_\theta(x) \geq 0.5$

predict "$y = 0$" if $h_\theta(x) < 0.5$

Let's use this to better understand how the hypothesis of logistic regression makes those predictions.

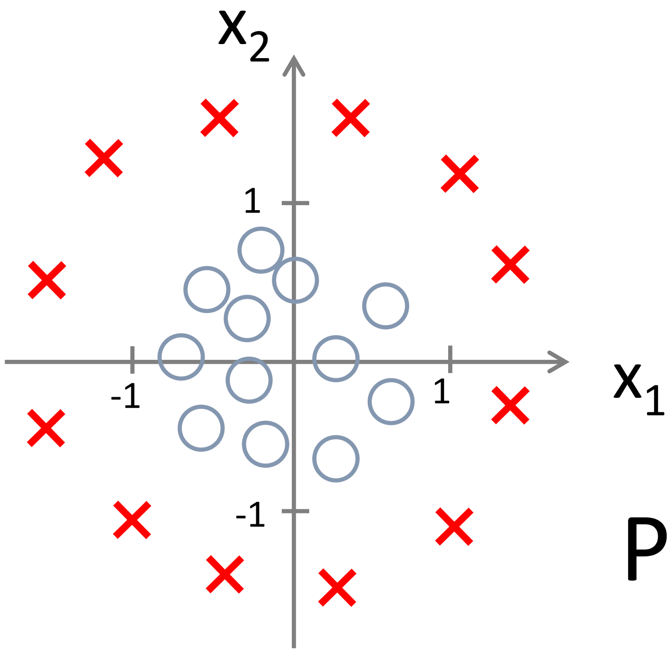
Decision Boundary



$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

Predict "$y = 1$" if $-3 + x_1 + x_2 \geq 0$

Non-linear decision boundaries



$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

Predict "$y = 1$" if $-1 + x_1^2 + x_2^2 \geq 0$

Now that we know what h(x) can represent.

Next: see how to automatically choose the parameters theta.
So that given a training set we can automatically fit the parameters to our data.

Logistic Regression

Cost Function

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \cdots, (x^{(m)}, y^{(m)})\}$

m examples $\quad x \in \begin{bmatrix} x_0 \\ x_1 \\ \cdots \\ x_n \end{bmatrix} \quad x_0 = 1, y \in \{0, 1\}$

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

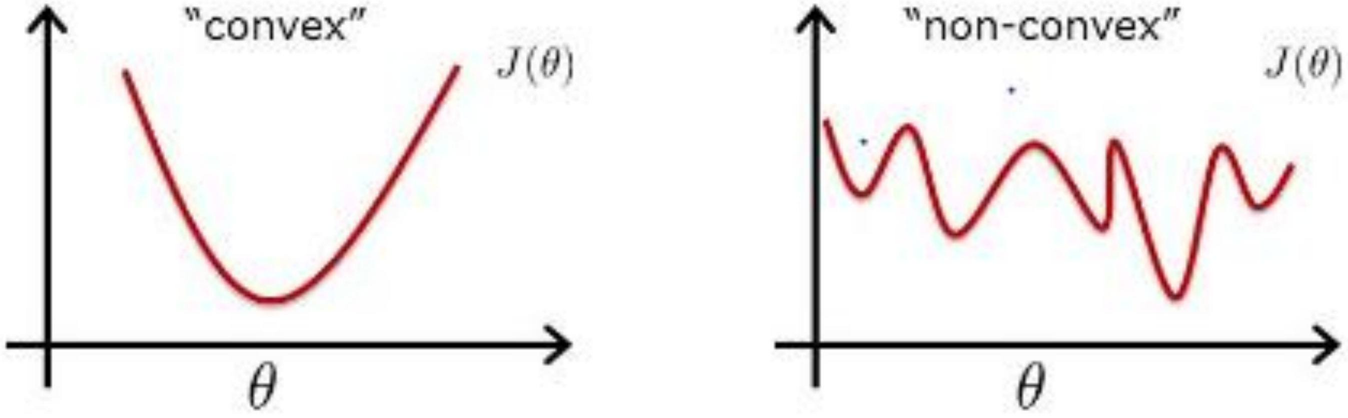
How to choose parameters θ ?

Cost function

Linear regression: $J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \frac{1}{2} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$

$$\mathrm{Cost}(h_\theta(x^{(i)}), y^{(i)}) = \frac{1}{2} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$$

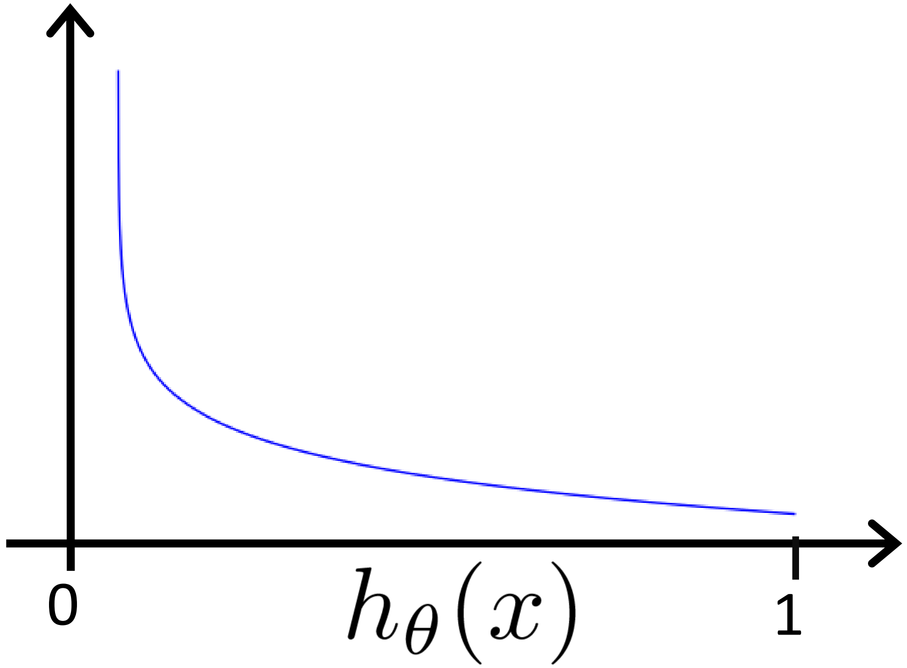
Convex Vs Non-Convex



Logistic regression cost function

$$\text{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

If y = 1



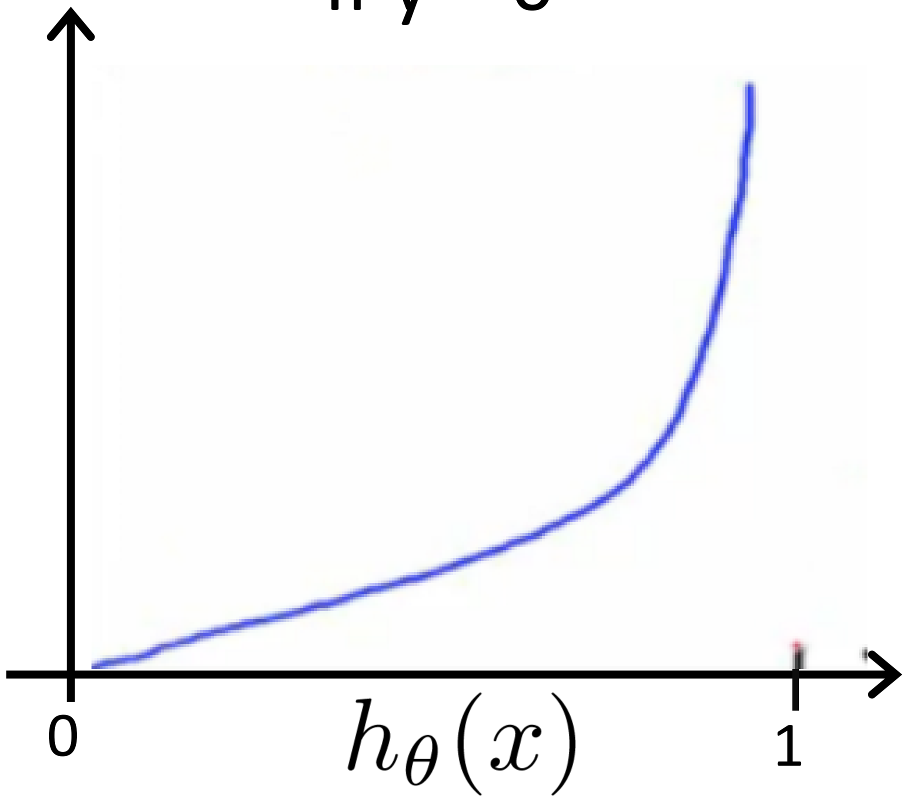
$\text{Cost} = 0 \text{ if } y = 1, h_\theta(x) = 1$
But as $h_\theta(x) \to 0$
$Cost \to \infty$

Captures intuition that if $h_\theta(x) = 0$, (predict $P(y = 1|x;\theta) = 0$), but $y = 1$, we'll penalize learning algorithm by a very large cost.

Logistic regression cost function

$$\text{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

If y = 0



Logistic Regression

Cost Function and Gradient Descent

Logistic regression cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \mathrm{Cost}(h_\theta(x^{(i)}), y^{(i)})$$

$$\mathrm{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

Note: $y = 0$ or 1 always

Logistic regression cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \mathrm{Cost}(h_\theta(x^{(i)}), y^{(i)})$$

$$= -\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log\left(1 - h_\theta(x^{(i)})\right)\right]$$

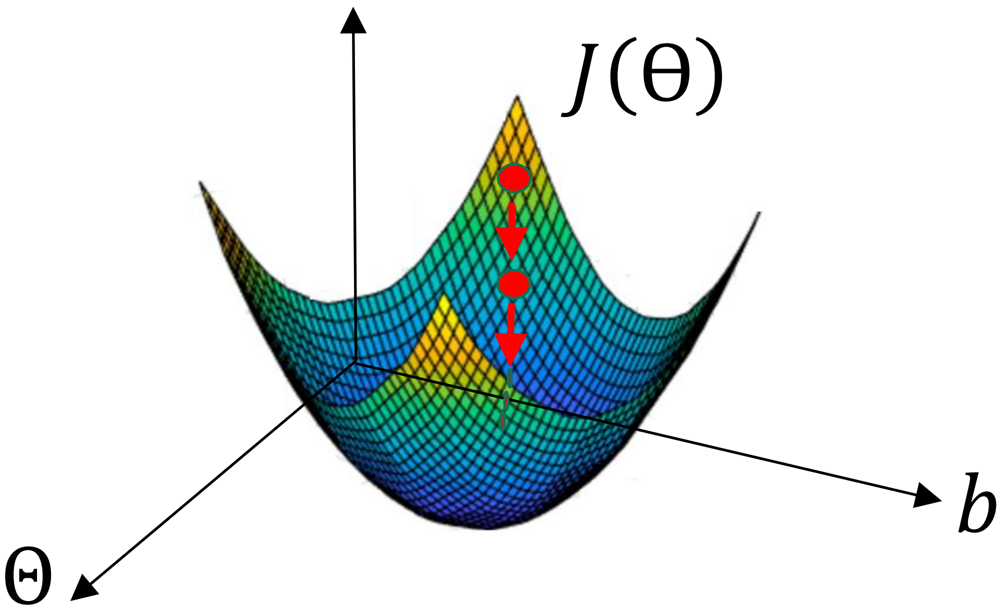
To fit parameters θ:

$$\min_\theta J(\theta)$$

To make a prediction given new x:

Output $h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}$

Gradient Descent



Gradient Descent

$$J(\theta) = -\frac{1}{m}[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_\theta(x^{(i)}))]$$

Want $\min_\theta J(\theta)$:

Repeat $\{$

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

(simultaneously update all θ_j)

$\}$

Gradient Descent

$$J(\theta) = -\frac{1}{m}[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_\theta(x^{(i)}))]$$

Want $\min_\theta J(\theta)$:

Repeat $\{$

$$\theta_j := \theta_j - \alpha \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

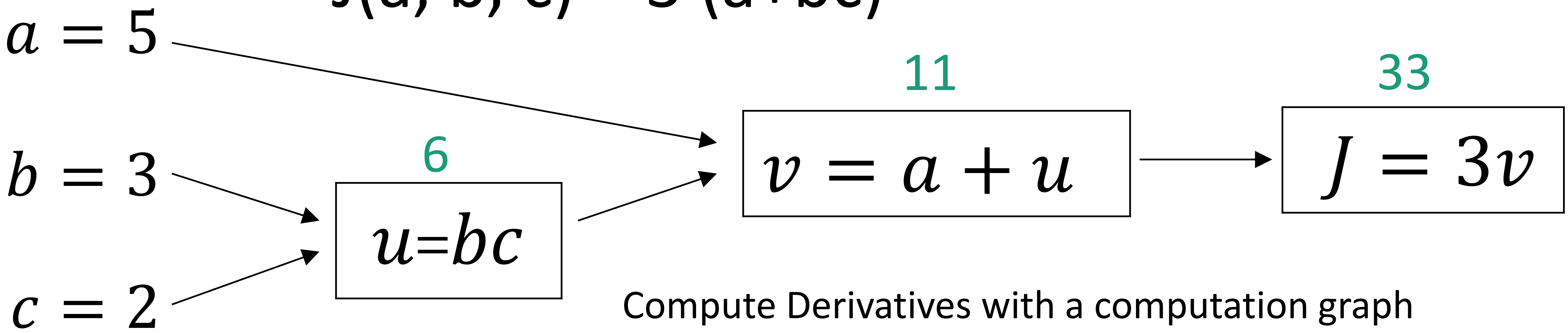
(simultaneously update all θ_j)

$\}$

Algorithm looks identical to linear regression!

Computation Graph

J(a, b, c) = 3 (a+bc)



Compute Derivatives with a computation graph

$$\frac{\partial J}{\partial u} = \frac{\partial J}{\partial v} * \frac{\partial v}{\partial u}$$

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial u} * \frac{\partial u}{\partial b}$$

$$\frac{\partial J}{\partial c} = \frac{\partial J}{\partial u} * \frac{\partial u}{\partial c}$$

$$\frac{\partial J}{\partial a} = \frac{\partial J}{\partial v} * \frac{\partial v}{\partial a}$$

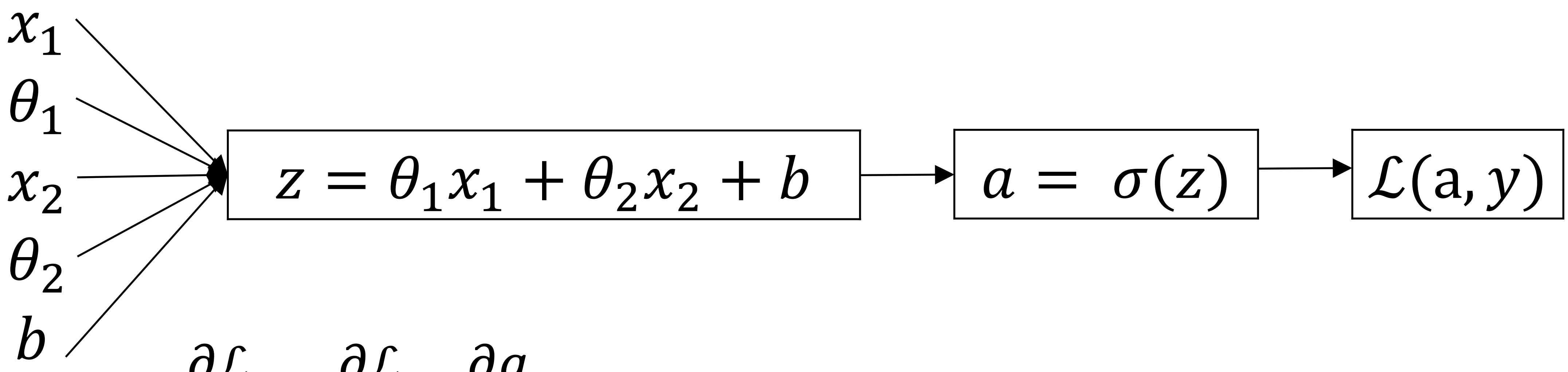
Computation Graph: Logistic Regression and Gradient

$$z = \theta^T x + b$$

$$\hat{y} = a = \sigma(z)$$

$$\mathcal{L}(a, y) = -(y \log(a) + (1 - y) \log(1 - a))$$

Logistic regression derivatives



$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial a} * \frac{\partial a}{\partial z} = a - y$$

$$\frac{\partial a}{\partial z} = a(1 - a)$$

$$\frac{\partial \mathcal{L}}{\partial a} = -\frac{y}{a} + \frac{1 - y}{1 - a}$$

Logistic regression on m examples

$$\mathrm{J}(\theta, b) = \frac{1}{m}\sum_{i=1}^{m} \mathcal{L}(a^i, y^i)$$

$$a^i = \hat{y} = \sigma\left(z^i\right) = \sigma(\theta^T x^i + b)$$

$$\frac{\partial}{\partial \theta_i} J(\theta, b) = \frac{1}{m}\sum_{i=1}^{m} \frac{\partial}{\partial \theta_i} \mathcal{L}(a^i, y^i)$$

Logistic regression on *m* examples

$J=0;\ d\Theta_1=0;\ d\Theta_2=0;\ db=0$

For i=1 to m

$z^i = \Theta^T x^i + b$

$a^i = \sigma(z^i)$

$J += -[y^i \log(a^i) + (1-y^i)\log(1-a^i)$

$dz^i = a^i - y^i$

$d\Theta_1 += x_1^i * dz^i$

$d\Theta_2 += x_2^i * dz^i$

$db += dz^i$

$J = J/m$

$d\Theta_1 /= m$

$d\Theta_2 /= m$

$db /= m$